

COMICCRAFT

AI-Based Comic Generation Application

Project Focus

Automatic generation of structured comic panels, narration, character dialogues, and comic-style illustrations from a user's story prompt.



Sample Comic Output – "The Last Star in the Sky"

Project Report

Prepared for academic project documentation and presentation.

1. Project Requirements

ComicCraft is designed as an AI-based comic generation application. The system accepts a custom story prompt from the user and converts it into a complete comic experience. The application must combine structured text generation with AI image generation so that each story can be presented as a sequence of visually rich comic panels.

Core Content Requirements

- **Structured comic outlines:** Generate the story panel-by-panel with a clear sequence and scene description.
- **Full narration and dialogues:** Produce narration, character conversations, emotions, and actions appropriate to each panel.
- **Comic-style illustrations:** Generate a visual illustration for every panel using detailed image prompts.
- **Creative and engaging output:** The generated comic should maintain story flow, character consistency, and visual richness.
- **Custom story support:** Users should be able to provide their own story idea, theme, characters, and setting.

Expected System Flow

1. User Prompt	2. Panel Outline	3. Narration & Dialogue	4. Image Prompt	5. Comic Panels
Custom story idea	Panel-by-panel structure	Detailed creative text	Visual scene description	Final illustrated comic

2. Model Evaluation

The selected AI technologies were evaluated according to the practical needs of ComicCraft. The evaluation focused on creative writing quality, prompt adherence, level of detail, generation speed, output quality, stability, documentation, and community support.

Technology	Main Purpose	Evaluation Focus
Gemini Flash	Fast structured text generation	Speed, structured panel outlines, prompt following
Gemini Pro	Detailed creative text generation	Narration quality, dialogue, creativity, scene detail
Stable Diffusion v1.5	AI image generation	Comic-style visuals, prompt adherence, image quality and flexibility

Evaluation Criteria

- **Response quality:** Ability to create natural narration and engaging character dialogues.
- **Prompt adherence:** Ability to follow story instructions and preserve important scene details.

- Richness of detail: Quality of descriptions for characters, locations, actions, mood, and visual composition.
- Speed: Practical response and generation time for an interactive application.
- Image generation quality: Ability to produce visually appealing comic-style illustrations from prompts.
- Stability and ecosystem: Documentation, implementation support, model availability, and community resources.

Practical Testing

Different story prompts and scene descriptions were used for practical testing. The text models were compared for their ability to create coherent panel structures, narration, and dialogue. The image-generation workflow was tested using scene prompts to observe comic-style visual quality, prompt adherence, and suitability for panel-based storytelling.

3. Final Model Selection

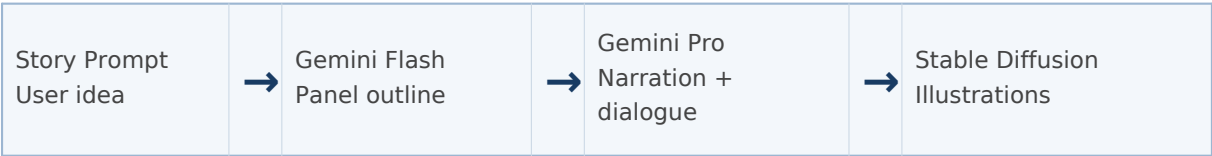
Based on the project requirements and practical testing, ComicCraft uses a combination of models rather than a single model. Each selected component is assigned a specific role so that the application can balance speed, creativity, structure, and visual generation.

Selected Model	Role in ComicCraft	Reason for Selection
Gemini Flash models/gemini-1.5-flash	Comic panel outline generation	Fast generation and suitability for structured, panel-by-panel outputs.
Gemini Pro models/gemini-1.5-pro	Narration and character dialogues	Designed for detailed and creative text generation with richer descriptions and dialogue.
Stable Diffusion v1.5 runwayml/stable-diffusion-v 1-5	Comic-style illustration generation	Provides flexible prompt-based image generation suitable for creating visual comic panels.

Why a Multi-Model Architecture?

- A fast model can handle structured planning without unnecessarily increasing generation time.
- A more detailed language model can focus on creative narration and character dialogue.
- A dedicated image-generation model can convert visual prompts into comic-style illustrations.
- Separating the tasks makes the overall pipeline easier to design, test, and improve.
- The combination supports an automated workflow from a simple user story to a complete illustrated comic.

ComicCraft Generation Pipeline



4. Sample Comic Output

The sample story used for demonstrating the ComicCraft concept is “The Last Star in the Sky.” It follows Mira, a curious girl who discovers a small glowing robot named Nova. The story demonstrates how a simple narrative idea can be transformed into multiple comic panels with narration, dialogue, and illustrations.



Figure 1: Generated sample comic – “The Last Star in the Sky”

Sample Story Structure

Panel	Title	Story Event
1	The Quiet Town	Mira watches the night sky.
2	A Strange Light	A bright star-like object falls into her garden.
3	The Little Visitor	Mira meets Nova, a small glowing robot.
4	The Search Begins	They follow blue sparks toward an old observatory.
5	The Hidden Crystal	Mira finds Nova’s lost energy crystal.
6	A Difficult Goodbye	Nova restores its power and prepares to return home.
7	The Promise	Nova gives Mira a glowing star-shaped device.
8	A New Star	Mira sees a new star and remembers her friend.

5. Resources

The following official documentation and model resources can be used for implementation, testing, and further development of ComicCraft.

Resource	Purpose
Google Gemini API Documentation	Documentation for integrating Gemini models and generating text-based outputs.
Google AI for Developers	Developer resources, examples, and guidance for generative AI applications.
Hugging Face Diffusers Documentation	Documentation for diffusion-based image generation pipelines and model usage.
Hugging Face Model Hub	Repository for discovering and using open-source models and related resources.
Stable Diffusion v1.5 Model Card	Model information and usage details for the selected Stable Diffusion checkpoint.

Conclusion

ComicCraft combines language models and image-generation technology to automate the creation of illustrated comics. The selected architecture divides the task into structured planning, detailed storytelling, and visual generation. This approach enables users to start with a simple custom story prompt and receive a complete comic consisting of organized panels, narration, dialogues, and illustrations. The proposed model combination provides a practical foundation for developing and demonstrating the ComicCraft application.

Project Name: ComicCraft

Application Type: AI-Based Comic Generation Application

Core Technologies: Gemini Flash, Gemini Pro, Stable Diffusion v1.5, Hugging Face Diffusers